

Chapter 58 — Management of Abnormalities of the Genitalia in Girls

FEMALE GENITAL EMBRYOLOGY

Cloacal division

- The **cloaca** is an endoderm-lined confluence of hindgut (dorsal) and allantois (ventral); it receives the **mesonephric ducts just before week 4**.
- The **urorectal septum** (appears week 4) separates the **urogenital sinus ventrally** from the **anorectal canal dorsally**. Two components:
 - **Tourneux fold** — coronal fold growing caudally from the allantois–hindgut angle towards the cloacal membrane.
 - **Rathke plicae** — lateral cloacal wall infoldings that coalesce in the midline and complete the septum caudally.
- By **weeks 6–7** the septum fuses with the cloacal membrane → **ventral urogenital membrane + dorsal anal membrane**.
- The resulting **fibromuscular node (perineal body)** is the insertion of the perineal muscles and divides the primitive cloacal sphincter into **urogenital diaphragm (anterior)** and **external anal sphincter (posterior)** — hence **both are supplied by the pudendal nerve**.

Mesonephric duct and ureteric bud

- Mesonephric ducts enter the urogenital sinus **near the müllerian tubercle**; the **ureteric bud** induces the **metanephric blastema** (secretory kidney).
- The terminal ureteric bud–mesonephric segment is absorbed into the sinus wall → **ureteric orifices at the lateral trigone**.

Müllerian ducts and the vagina

- **Paramesonephric (müllerian) ducts** form from coelomic epithelium **lateral to the mesonephric ducts**, cross medially and fuse → **proximity explains paramesonephric anomaly + ipsilateral renal anomaly**.
- Caudal fusion with **dissolution of the midline partition** → common **uterovaginal canal** → **uterus, cervix, proximal two-thirds of vagina**. **Failure of septal regression** → müllerian duct anomalies.
- **Koff (1933)**: contact of the uterovaginal primordium with the sinus forms the **müllerian tubercle**, which induces paired endodermal **sinovaginal bulbs** (possibly terminal wolffian segments).
- Sequence: sinovaginal bulbs proliferate → solid cord → **vaginal plate** → **canalised caudal-to-cranial** → distal vagina.
- Sinus distal to the tubercle **everts (exstrophies) to form the vestibule** → separate urethral and vaginal openings.
- **Hymen** = invagination of the posterior wall of the urogenital sinus separating vaginal lumen from sinus; normally **ruptures perinatally**.

Vestibular glands and cystic remnants

- **Skene (paraurethral) glands** — homologues of the prostatic glands; **Gartner duct** — wolffian duct remnant; **Bartholin (greater vestibular) glands** — urogenital sinus outgrowths, homologues of the **bulbourethral glands**.

EMBRYOLOGIC ORIGINS OF FEMALE GENITAL STRUCTURES

Primordium	Derivative
Müllerian (paramesonephric) ducts	Uterus, cervix, proximal two-thirds of vagina
Sinovaginal bulbs → vaginal plate	Distal vagina
Urogenital sinus distal to müllerian tubercle	Vestibule ; hymen (posterior wall invagination)
Wolffian (mesonephric) duct	Ureteric bud; remnant = Gartner duct
Urogenital sinus glands	Skene (≈ prostate); Bartholin (≈ bulbourethral)
Cloacal septation node	Perineal body; pudendal-innervated sphincter complexes

- The table is testing **which primordium fails in which anomaly**: müllerian failure removes uterus and upper vagina; sinus failure removes only the distal vagina.

Skeletal development and caudal regression

- Vertebrae form **craniocaudally**; lower limb buds from **somites 25–29**, differentiating **weeks 4–8**.
- A **segmentation disturbance at the caudal somites when the fetus is <10 mm (weeks 4–5)** affects many systems → **caudal regression syndrome (Duhamel, 1961)**: terminal bowel, kidney, bladder, müllerian system, lumbosacral spine.
- Proposed mechanisms: **disordered mesodermal migration, reduced proliferation, premature apoptosis**; increased in **infants of diabetic mothers**; **homeobox gene deletions** reproduce the constellation.
- Because somites differentiate cranial-to-caudal, **earlier insults → higher anorectal malformations** with more severe upper tract, genital duct and spinal anomalies than low imperforate anus.
- Mesodermal disturbance can extend cranially to **C1 and the tracheo-oesophageal anlagen (VATER, MURCS)**.

MESODERMAL MALFORMATION ASSOCIATIONS

Association	Components	Distinguishing feature
Caudal regression (Duhamel)	Terminal bowel, kidney, bladder, müllerian ducts, lumbosacral spine	Caudal somite insult weeks 4–5; maternal diabetes
VATER	Vertebral defects, Anal atresia, Tracheo-oesophageal fistula with oesophageal atresia, Radial and renal dysplasia	Extends to tracheo-oesophageal anlagen
MURCS	MÜllerian duct aplasia, Renal aplasia, Cervicothoracic Somite dysplasia	Cervical (Klippel-Feil) level; MRKH type B

KEY POINTS: FEMALE GENITAL EMBRYOLOGY

- Common ontogeny of urogenital and external anal sphincters → **pudendal nerve supplies both**.
- Müllerian–wolffian proximity → **paramesonephric anomalies with ipsilateral renal anomalies**.
- Caudal somite segmentation disturbance in **weeks 4–5** → urogenital, skeletal and gastrointestinal malformations = **caudal regression syndrome**.

Concept — Genital, renal, anorectal and spinal anomalies travel together because they are **built from adjacent tissue in the same week**; the earlier and more cranial the insult, the more systems it takes with it.

EVALUATION AND CLASSIFICATION OF FEMALE GENITAL ANOMALIES

Examination

- Differential of vulval anomalies is broad; **age and ancestry narrow it, but physical examination is the most useful tool.**
- **Reassure the child that examination is painless; frog-leg position.** Record **clitoral size, hymenal configuration, urethral location, character of any interlabial mass** (smooth, lobulated, haemorrhagic).
- **Pull-down manoeuvre:** labia majora grasped and drawn **caudally and laterally** → funnels introitus, displays distal vagina.
- Define the mass relative to landmarks: **lubricated cotton applicator posteriorly** (vagina) and/or **small feeding tube in the suspected urethra.**
- Awake inspection: otoscope, nasal speculum or paediatric vaginal speculum. **Vaginal discharge or bleeding** → **examination and vaginoscopy under anaesthesia.**
- **Renal–pelvic ultrasonography** is a useful adjunct in selected disorders.

Classification and imaging

THREE CATEGORIES OF VAGINAL STRUCTURAL ANOMALY

Category	Mechanism	Examples
1. Hypoplasia / agenesis	Structure fails to form	Vaginal agenesis (Mayer-Rokitansky-Küster-Hauser [MRKH] syndrome)
2. Vertical fusion	Abnormal müllerian–urogenital sinus contact (canalisation)	Imperforate hymen, transverse septum, distal vaginal atresia
3. Lateral fusion	Failed midline fusion/resorption (duplication)	Didelphys, Herlyn-Werner-Wunderlich syndrome (HWWS)

- **Ultrasonography:** genital anatomy and screening for upper tract anomalies.
- **Magnetic resonance imaging (MRI)** = gold standard for müllerian anatomy — especially **presence of a cervix and functioning endometrium.**
- Complex cases: **examination under anaesthesia, vaginoscopy, hysteroscopy, laparoscopy.**
- **Obstructive anomalies** → **immediate intervention; non-obstructive** → **usually no surgery** until reproductive age and only if intercourse or fertility is affected.
- Most inclusive classification: **American Society for Reproductive Medicine (ASRM).**

Concept — The operative decision turns on **one question: is it obstructive?** Obstruction mandates prompt drainage; everything else waits for function to demand it.

CONGENITAL DISORDERS OF FEMALE EXTERNAL GENITALIA

Disorders of the Clitoris

Hypertrophied Clitoris

- Commonest cause: **congenital adrenal hyperplasia (CAH)** — androgenic metabolites from an adrenal steroid enzyme block; **21-hydroxylase or 11 β -hydroxylase deficiency.** Some clitoromegaly in nearly every girl with CAH.
- Degree of clitoromegaly **often but not always** matches common urogenital sinus length.
- **Any newborn with clitoromegaly: serum electrolytes, 17-hydroxyprogesterone, karyotype.**
- **Glucocorticoid + mineralocorticoid replacement** → negative feedback removes further androgenic stimulation.

- **Palpate the corporal bodies:** separates true enlargement from **prominent clitoral hood skin** (possible lymphatic drainage abnormality).

CAUSES OF CLITORAL ENLARGEMENT

Cause	Mechanism	Clue
CAH (commonest)	Adrenal androgen excess	17-hydroxyprogesterone, electrolytes
Maternal androgen-secreting tumour	Transplacental androgen	Maternal history
Turner mosaic disorder of sexual development (DSD) with viable testicular tissue	Gonadal androgen (unilateral)	Karyotype
Plexiform neurofibroma (neurofibromatosis)	Local growth factor secretion	Neurofibromatosis stigmata
Pseudo-enlargement (hood skin)	Lymphatic abnormality	Normal corpora on palpation

- The table's trap: **not all enlargement is androgenic**, and not all apparent enlargement is clitoral.

Diminutive Clitoris

- **Complete androgen insensitivity syndrome (CAIS)** → reduced clitoral length versus controls; **clitoral-to-urethral distance unchanged**.

Disorders of the Vestibule

Urethral Prolapse

- **Complete circumferential eversion of urethral mucosa** at the meatus; first described by **Solinger (1732)**.
- Bimodal: **prepubertal girls of African ancestry** and **postmenopausal White women**.
- Proposed causes: **hypo-oestrogenism**, abnormal connections between **inner longitudinal and outer circular** distal urethral muscle, **episodic raised intra-abdominal pressure**.
- Commonest complaint: **bleeding / spotting on underwear** from oedematous friable mucosa.
- Sign: **doughnut-shaped mass with meatus at centre**; if a catheter passes through the centre, **no imaging is needed**.
- Treatment: observation, topical corticosteroid, conjugated oestrogen, sitz baths → spontaneous reduction possible but **recurrence after non-operative treatment is high**.
- **Procedure of choice:** circumferential excision of redundant mucosa + suture of normal urethra to vestibule. **Avoid ligation over a catheter (sloughing) and cryosurgery**.

Urethral Polyp

- **Paediatric equivalent of the urethral caruncle**; rare; presents as an **interlabial mass**.
- Likely **hamartomatous or inflammatory**; benign, **transitional and squamous epithelium**. Include in the differential of any interlabial mass.

Vestibular Cysts

- Three neonatal introital cysts: **paraurethral (Skene)**, **Gartner duct**, **covered ectopic ureter**. **Large cyst emanating from the vaginal introitus** → renal-pelvic ultrasonography for duplication.
- **Paraurethral cyst:** dilated periurethral gland (6–30 glands; two largest = Skene) responding to **maternal oestrogen** with mucoid secretion. Hallmark: **displaced meatus** → **eccentric stream**. Meatus clearly separate → **no imaging**. Usually **self-limited, ruptures**; persistent → **bedside needle drainage**.
- **Gartner duct cyst:** incomplete wolffian regression; **anteromedial vaginal wall**.

- **Covered ectopic ureter:** an ectopic ureter may end in a wolffian segment (= Gartner duct); usually ruptures before birth into the vagina; if epithelium persists → **urine-filled covered cyst**. **Intraoperative contrast** defines anatomy; **incision relieves obstruction**.
- **More ectopic insertion → more dysplastic segment**, yet vaginal-inserting moieties may still make urine → **upper pole heminephrectomy if significant urine output (incontinence) or infection after decompression**.

Prolapsed Ureterocele

- Ectopic ureterocele = cystic dilatation of the terminal ureter; **~90% subtend the upper pole of a duplex system**.
- Prolapse through the urethra during voiding, **almost always in infancy → urinary retention; relative urological emergency**. Colour **pink → dusky purple** with duration.
- **Bladder–renal ultrasonography: upper pole hydronephrosis in a duplex system**.
- Sequence: **needle decompression or incision → reduction → urethral catheter**; general anaesthesia often preferable (infant straining); contrast injection aids anatomy.

Ectopic Ureteral Insertion

- In girls the ectopic ureter may enter **distal to the urethral sphincter → diurnal incontinence despite normal voiding**.
- Usually **upper pole of a duplex**; single-system ectopia occurs and may mimic **absent kidney → search for an ectopic kidney**.

Labial Enlargement

- Imaging: **ultrasonography first, Doppler** for vascularity, **MRI** as adjunct.
- **Commonest cause: inguinal hernia** — usually patent processus vaginalis, but **consider CAIS**: a large proportion of CAIS girls undergo childhood hernia repair, and CAIS is found in a small but real fraction (~1–2.5%) of girls with hernia.
- Excluding CAIS: **uterus on preoperative ultrasonography, karyotype, or vaginoscopic cervix at surgery; prenatal cell-free DNA without a Y chromosome** confirms 46,XX and may make these unnecessary.
- If CAIS: **preserve the testis** (oestrogen via peripheral aromatisation of testosterone); place intra-abdominally or **superficial to the fascia** for easier monitoring.
- **Hernia + phallic enlargement** → mosaic DSD, ovotesticular DSD, partial androgen insensitivity syndrome, **5 α -reductase deficiency**.
- **Childhood asymmetric labium majus enlargement (CALME)**: physiological expansion of normal vulval tissue, **no capsule; soft, fleshy, painless**; fibrous tissue enclosing fat lobules; oestrogen receptor-positive, but low prepubertal oestrogen questions a hormonal role.
- **No well-delineated mass → observe; continued growth → other aetiologies**: plexiform neurofibroma, sarcoma, vascular/lymphatic malformation, malignant peripheral nerve sheath tumour, **metastatic (extraintestinal) Crohn disease**. **Well-demarcated mass → consider excision**.

DIFFERENTIAL DIAGNOSIS OF THE INTERLABIAL / INTROITAL MASS IN GIRLS

Entity	Typical patient	Distinguishing feature	Confirmation	Management
Urethral prolapse	Prepubertal, African ancestry	Doughnut, meatus central ; spotting	Catheter through centre; no imaging	Conservative; circumferential excision
Urethral polyp	Young child	Rare; caruncle equivalent	Histology	Excision

Entity	Typical patient	Distinguishing feature	Confirmation	Management
Paraurethral (Skene) cyst	Neonate	Meatus displaced, eccentric stream	Separate meatus; no imaging	Observe; needle drainage
Gartner cyst / covered ectopic ureter	Neonate	Anteromedial vaginal wall; large introital cyst	Renal–pelvic ultrasonography; contrast	Incision; heminephrectomy if needed
Prolapsed ureterocele	Infant	Pink–purple, urinary retention	Upper pole hydronephrosis	Decompress, reduce, catheter
Imperforate hymen	Neonate / adolescent	Whitish (neonate) / bluish (adolescent) bulge	Clinical ± suprapubic mass	Transverse incision
Rhabdomyosarcoma	Mean age <2 years	Grapelike cluster, bleeding	Biopsy; staging	Chemotherapy ± local resection
Hymenal skin tag	Any	Rarely symptomatic	—	Excise if bleeding

- What the table tests: **locate the meatus first** — central (prolapse), displaced (Skene cyst), separate and vaginal (Gartner/ectopic ureter, hymen, tumour). **Any vaginal-origin cyst mandates renal imaging.**

KEY POINTS: DISORDERS OF THE CLITORIS, VESTIBULE, AND URETHRA

- **Diminutive clitoris → CAIS.**
- Introital (interlabial) mass associated with the urethra → **always renal–pelvic ultrasonography.**

Figure legends

- **Fig. 58.4** — Clitoral hypertrophy in Turner syndrome with **mosaic DSD and functional testicular tissue.**
- **Fig. 58.5** — Clitoral hood skin enlargement; **normal-size corpora and glans on palpation.**
- **Fig. 58.8** — Urethral polyp: probe in urethra, Babcock clamp on polyp.
- **Fig. 58.10** — Retrograde injection of a Gartner duct cyst: contrast ascends into an **atretic, dysplastic ureter.**

Concept — Every vestibular mass is read **against the urethral meatus and the kidney**: its relation to the meatus names it, and a wolffian or ureteric origin means the upper tract must be imaged.

Disorders of the Vagina

Imperforate Hymen and Hymenal Skin Tags

- **Hymenal skin tags**: almost always normal; **excise if symptomatic (bleeding)** to exclude malignancy.
- Congenital hymenal variants range from **imperforate** to **microperforate**.
- **Imperforate hymen = probably the commonest congenital obstructive anomaly** of the female tract.

IMPERFORATE HYMEN: NEONATAL VERSUS ADOLESCENT PRESENTATION

Neonatal presentation	Adolescent presentation
Bulge at posterior introitus / suprapubic mass; whitish (mucocolpos from maternal oestradiol)	Primary amenorrhoea ± cyclic pain; bluish bulge; mass on rectoabdominal palpation
After neonatal period mucus reabsorbs → bulge may vanish	Haematocolpos

- Newborn: **bedside incision in the transverse direction** — avoids extension anteriorly (urethra) or posteriorly (rectum).
- **Simple aspiration without a drainage procedure is discouraged** (residual material → ascending infection). Pubertal: excision of excess tissue under general anaesthesia.

- **Acquired hymenal abnormalities usually = sexual abuse:** haematoma, abrasion, laceration with transection → **examination under anaesthesia + forensic swabbing by standardised protocol.**

Abnormalities of Vertical Fusion

General

- Present **after puberty** with **amenorrhoea, abdominopelvic pain**, distension mimicking pregnancy. **Delay** → **retrograde menstruation** → **endometriosis.**

Transverse vaginal septum

- Failure of **fusion and/or canalisation** of urogenital sinus and müllerian ducts; **~1 in 70,000 females.**
- Post-menarche: **amenorrhoea, pelvic pain, urinary retention** from haematocolpos; neonates/children: **mucocolpos**; or incidental.
- Level: **upper and middle third commonest, lower third least.** Usually **<1 cm thick**, often a small central/eccentric perforation — **obstruction and ascending infection still occur despite perforation.**
- Diagnosis: **transperineal, transrectal, abdominal ultrasonography and MRI** (level, thickness). **MRI distinguishes a high septum from cervical agenesis** — misdiagnosis causes serious morbidity.
- If pain is controllable: **defer surgery with gonadotropin-releasing hormone (GnRH) agonist or continuous oral contraceptive** → lower segment dilates, easing repair.
- Techniques: **simple incision; excision with approximation of upper and lower mucosal edges; Z-plasties.**
- **Balloon-assisted division:** ultrasound-guided **14-gauge angiocatheter** through septum → aspirate thick blood → **0.035-inch guidewire** → **NephroMax balloon** dilatation → excise remaining septum; thick septum → resectoscope loop excision.
- **Transabdominal approach:** posterior wall of distended vagina opened vertically → septum perforated → **Foley balloon left 2 weeks**; used where sociocultural belief requires an **intact hymen.**
- **Commonest complication: stenosis at resection site.** **Z-plasty** converts a contracting transverse line into a **longitudinal one**; **postoperative mould** further reduces stenosis.

POSTOPERATIVE MOULD: HOLLOW VERSUS SOLID

Mould after septum resection	Mould after vaginoplasty (e.g., McIndoe)
Hollow — uterus present; menstrual egress	Solid — no uterus

TRANSVERSE SEPTUM: EARLY VERSUS DEFERRED CORRECTION

Early correction (neonate/child)	Deferred correction
May reduce endometriosis	Larger anatomy , better technical feasibility; better dilation compliance

Vaginal atresia (distal vagina)

- **Urogenital sinus fails to form the lower vagina; müllerian structures normal** → uterus, cervix, upper vagina present (distinct from vaginal agenesis and CAIS).
- Examination: **very shallow dimple** below the urethra; **distended vagina palpable rectally** distinguishes it from CAIS or agenesis.
- **Ultrasonography and/or MRI mandatory before intervention.**
- Sequence: **transverse incision at hymenal ring** → **dissect through fibrous zone** → **reach distended upper vagina** → **drain** → **pull-through to introitus.** Haematocolpos acts as a **tissue expander.** Gap bridged by **perineal skin flaps and/or vaginal mobilisation**; **combined abdominoperineal approach** aids exposure and anastomosis.

Vaginal agenesis (müllerian aplasia)

- **~1 in 5000 live female births: congenital absence of the proximal vagina in a phenotypically, chromosomally (46,XX) and hormonally (normal luteinising hormone and follicle-stimulating hormone) intact female.**
- **MRKH syndrome: sinovaginal bulbs fail to develop and form the vaginal plate**, possibly from failed induction by the uterovaginal primordium; coincident mesodermal organogenesis explains associated findings. Associated with **maternal galactose-1-phosphate uridylyltransferase deficiency**.
- **Hymenal fringe and small vaginal pouch present** (urogenital sinus-derived) — unlike vaginal atresia.
- Presents **after expected menarche: second commonest cause of primary amenorrhoea after gonadal dysgenesis**; minority have **cyclic pain** (functioning uterine remnant).
- **Inguinal hernia less common than in CAIS**. Imaging: remnant müllerian structures, **renal and skeletal anomalies**.

Associated Findings

- Variable absence/hypoplasia of cervix, uterus and tubes; **~10% have an obstructed normal or rudimentary uterus with functioning endometrium**. Most lack a uterus or have a solid rudimentary one; tubes frequently rudimentary.
- **Ovaries always present and functional** (occasionally cystic).
- **Renal anomalies in about one-third** (first noted by **Rokitansky, 1838**) — **almost exclusively type B**; mostly **unilateral renal agenesis or ectopia**. Converse holds: girls with renal anomalies frequently have genital anomalies.
- **Skeletal anomalies 10–20%**. **Klippel-Feil** (cervical vertebral fusion; ~1 in 30,000–40,000 births) → **MURCS association (Duncan, 1977)**, generalised mesodermal defect in week 4; **Klippel-Feil only in type B**. Also scoliosis, hand and face anomalies.
- **Abnormal pelvic vasculature** — awareness prevents surgical misadventure.
- **Not associated with lumbosacral spinal disorders or occult dysraphism** (unlike anomalies with abnormal cloacal septation).
- Type B proportion **varies with ethnicity** (high in White cohorts, lower in Chinese populations).

MRKH SYNDROME: TYPICAL (TYPE A) VERSUS ATYPICAL (TYPE B)

	Type A (typical)	Type B (atypical)
Uterine remnants	Symmetrical	Asymmetrical uterine buds
Fallopian tubes	Normal	Abnormally developed
Renal anomalies	Essentially absent	Common
Klippel-Feil / skeletal / MURCS	Absent	Present

- The classification exists because **type predicts the associated anomaly burden**: type B warrants full renal and skeletal work-up.

KEY POINTS: DISORDERS OF THE VAGINA

- Three categories: **hypoplasia/agenesis, vertical fusion, lateral fusion**.
- Transverse septum: failure of **fusion or canalisation** of urogenital sinus and müllerian ducts.
- Vaginal atresia: **urogenital sinus fails** to form the distal vagina.
- Vaginal agenesis: absent proximal vagina in a **phenotypically, chromosomally, hormonally intact** female.

- **MRKH: sinovaginal bulbs fail to develop and form the vaginal plate.**

OBSTRUCTED OR ABSENT VAGINA — COMPARISON

Entity	Level of failure	Uterus / cervix	Clue; key investigation	Treatment
Imperforate hymen	Hymen	Normal	Bulging membrane at introitus; Clinical	Transverse incision
Transverse septum	Sinus-müllerian junction; upper/middle third	Normal	Normal hymen, blind vagina above; Ultrasonography, MRI (cervix?)	Excision, Z-plasty, balloon; hollow mould
Vaginal atresia	Urogenital sinus (distal vagina)	Normal	Shallow dimple; distended vagina per rectum ; Ultrasonography/MRI mandatory	Pull-through ± flaps
MRKH	Sinovaginal bulbs / vaginal plate	Absent or rudimentary	Hymenal fringe + small pouch ; 46,XX; ovaries normal; Imaging for renal/skeletal	Dilation first line; vaginoplasty
CAIS	Androgen resistance (46,XY)	Absent	Short vagina, hernia, diminutive clitoris; Karyotype	Preserve gonad
Cervical agenesis	Cervix	Uterus present, no cervix	Obstructive symptoms; MRI	Often hysterectomy + neovagina
HWWS	Lateral fusion + unilateral obstruction	Didelphys	Menstruates ; lateral bluish vaginal bulge; Ultrasonography, MRI	Wide incision of vertical septum

- The pattern: **where the uterus is present, the problem is drainage**; where it is absent, the problem is a vagina. The trap is the **high septum mistaken for cervical agenesis** — only MRI separates them.

Vaginal Replacement Surgery

Creation of a skin neovagina

- Helpful for the patient to **speak with someone previously treated** before choosing.
- **Frank (1938) dilation**: graduated hard dilators pressed on the perineum → progressive invagination of the dimple; **Ingram modification = bicycle-seat stool**. Best when a **dimple or pouch exists**.
- **Progressive perineal dilation = first-line therapy** in adolescents; a multicentre study found **surgery not superior to dilation** → **non-surgical options always before vaginoplasty**.
- **Vecchietti**: Plexiglas “olive” against the dimple, traction sutures passed cranially through vesicorectal space and abdominal wall and progressively tensioned; **laparoscopic assistance** avoids laparotomy; healthy mucosa, near-normal length at 3 months; **little used by urologists**.
- **Abbe (1898)**, the first landmark: canal between rectum and urethra lined with **split-thickness skin** → popularised as the **McIndoe** procedure.
- **McIndoe sequence**: **full mechanical + antibiotic bowel preparation** → **split-thickness buttock graft (0.018–0.022 inch)** tubularised over stent → **transverse incision at dimple** → **dissect urethrorectal space to peritoneal reflection** → **insert graft and mould** → **labia minora sutured over stent** → **suprapubic catheter** → **strict bed rest 1 week**.
- Stents: packed gauze, condom-covered wood, silicone foam, acrylic, metal, inflatable. **High postoperative stenosis** → **dilation required**; satisfaction excellent.
- **Full-thickness graft** (buttock) or labial flaps → **less contracture** than split-thickness. **Williams (1964)**: labia majora pouch ± Frank dilation.

- **Buccal mucosa:** hairless, moist, **non-keratinised stratified squamous**; dense elastic fibres, **robust neovascularisation**; also for revision and partial replacement after tumour excision; favourable long-term results.
- Others: **muscle flaps** (pudendal thigh, rectus abdominis, buttock), **pelvic peritoneum**, **amnion**.
- **Robotic-assisted tubularised peritoneal vaginoplasty (RATPV):** single peritoneal flap tubularised and joined to inverted non-hair-bearing perineal skin; good length and calibre; **no bowel harvest, no laser hair removal**; used mainly in transgender surgery, promising for agenesis.
- **Autologous cultured vaginal tissue** with McIndoe technique: normal length and depth, satisfactory sexual function.

Creation of an intestinal neovagina

- **Baldwin (1907):** U-shaped sigmoid loop with septum division; abandoned for morbidity and mortality until improvements from the 1970s. **Sigmoid, caecum, small bowel** all usable.
- Sigmoid sequence: **bowel preparation → frog-leg (Allen stirrups if older) → select distal sigmoid on a pedicle reaching perineum → divide distal end, anastomose to perineum → excise ~3 cm intervening segment** (no overlapping suture lines → fewer fistulae) → **fix to posterior peritoneum** (prevents prolapse).
- **Modified Monti detubularisation:** open along one side near mesentery, retubularise **Heineke-Mikulicz** → mesentery cranial, caudal end reaches further.
- **Intact cervix → bowel can be anastomosed to the vagina**, allowing menstrual egress.
- Ileum: detubularised, reconfigured conically for diameter; **laparoscopic-assisted** harvest avoids laparotomy.
- **Perineal dimple present → anastomose bowel directly to it**; otherwise create an adequate rectovesical space (**progressive Hegar dilators**) to protect blood supply; pass segment inside a **lubricated Penrose drain**.
- Results excellent: **mucosal inversion → simple trimming**; **mild stenosis → Z-plasty** at mucocutaneous junction; most achieve adequate coitus.

LARGE BOWEL VERSUS ILEUM FOR INTESTINAL NEOVAGINA

Sigmoid / large bowel (choice)	Ileum
Wider lumen, mobile mesentery → less stenosis	More stenosis (narrow lumen, short mesentery)
Default segment	Only for prior deep pelvic irradiation or absent large bowel (cloacal exstrophy)

BOWEL VERSUS MCINDOE SKIN NEOVAGINA

Bowel neovagina	McIndoe skin neovagina
Lubrication; less contracture → less dilation	More stenosis → mandatory dilation
Chronic discharge → pads, douching; higher blood-borne pathogen transmission (poor barrier)	Squamous barrier
Adenocarcinoma reported	Condylomata, squamous cell carcinoma reported

- **All neovaginas → annual examination.**

TIMING OF VAGINAL REPLACEMENT

Defer to adulthood	Do not defer
McIndoe advocates — maturity needed for daily dilation	Bowel advocates — delay may be psychologically traumatic

- Author's practice: timing by **underlying diagnosis and need**.

NEOVAGINA TECHNIQUES — COMPARISON

Technique	Principle	Strength	Weakness
Frank / Ingram dilation	Progressive perineal pressure	First line ; non-inferior to surgery	Needs dimple and compliance
Vecchietti	Olive + abdominal traction (laparoscopic)	Rapid length, healthy mucosa	Little urological use
McIndoe (Abbe)	Split-thickness skin over stent	Widely accepted, high satisfaction	Stenosis ; bed rest; squamous cell carcinoma risk
Full-thickness graft / Williams	Buttock graft; labial pouch	Less contracture	—
Buccal mucosa	Non-keratinised oral mucosa	Vascularity, graft take; revisions	—
RATPV	Tubularised peritoneal flap	No bowel harvest or hair removal	Limited experience in agenesis
Sigmoid	Pedicled colon	Lubrication, less contracture	Discharge, adenocarcinoma
Ileum	Pedicled small bowel	Irradiated pelvis, no colon	More stenosis

- The organising trade-off: **skin stenoses, bowel secretes**; newer tissues (buccal, peritoneum) aim to avoid both.

KEY POINTS: VAGINAL REPLACEMENT SURGERY

- **McIndoe** → **more stenosis** than bowel neovagina.
- Intestinal neovagina stenosis: **ileum > large bowel**.
- **Cervical agenesis: avoid perineum-to-uterus reconstruction** — life-threatening ascending infection.

Special Considerations: Cervical Atresia

- Uncommon; mimics other obstruction (**primary amenorrhoea, cyclic or chronic pain**). Wrong diagnosis/operation → disaster, possible death.
- **MRI = most useful modality**.
- **Hysterectomy + vaginal replacement** often best.
- **Uterovaginal anastomosis** can succeed with pregnancies (caesarean delivery), but **loss of the endocervical barrier** → **life-threatening ascending infection**.
- Continuity impossible → **in vitro fertilisation with transmyometrial embryo transfer** (caesarean delivery); meanwhile **hormonal endometrial blockade** to limit pain and endometriosis.

Abnormalities of Lateral Fusion

- Most **functionally insignificant, asymptomatic** — exceptions: **uterus didelphys and unilateral vaginal obstruction**.
- **True uterine duplication**: rare; duplication of the müllerian ducts.

- **Uterus didelphys** (far commoner): **failed resorption of the medial wall** → two cavities, two cervices; **septate vagina in up to 75%**; reproductive outcome usually adequate.
- **Excise vaginal septum only for dyspareunia, obstructed vaginal delivery or need for two tampons.**
- Persistence of only the cranial part of the uterine septum gives a **septate uterus** with normal vagina; **septum incision only for recurrent pregnancy loss**. (A **bicornuate** uterus results from incomplete fusion, not septal persistence.)

Duplication of the uterus and cervix with unilaterally imperforate vagina (Herlyn-Werner-Wunderlich syndrome)

- **HWWS**: uterus didelphys + unilateral imperforate hemivagina — obstruction arising from a **lateral** fusion anomaly.
- Cyclic/chronic pain but **no primary amenorrhoea** (contralateral side menstruates).
- Sign: **unilateral abdominopelvic mass ending in a bluish lateral vaginal wall bulge**. Ultrasonography and MRI define anatomy.
- **Ipsilateral renal anomaly common — renal agenesis commonest.**
- Prompt diagnosis prevents damage from **cryptomenorrhoea and endometriosis**.
- **Treatment: wide incision of the vertical vaginal septum.**

HWWS: ZHU CLASS 1 VERSUS CLASS 2

Zhu Class 1	Zhu Class 2
Complete hemivaginal obstruction	Incomplete — drains via small perforation
Earlier presentation	Later presentation, less endometriosis ; the commoner class

LATERAL FUSION ANOMALIES

Anomaly	Mechanism	Vagina	Intervention
True duplication	Müllerian duct duplication	Duplicated	Rare
Uterus didelphys	Failed medial wall resorption	Septate in most	Vaginal septum excision if symptomatic
Septate uterus	Cranial septum persists	Normal	Incision for recurrent loss only
HWWS	Didelphys + unilateral obstruction	Obstructed hemivagina	Wide septal incision

Figure legends

- **Fig. 58.18** — Transverse septum levels: A high (upper vagina), B middle, C low.
- **Fig. 58.19** — Balloon-assisted division: haematocolpos on MRI → ultrasound-guided angiocatheter → NephroMax balloon.
- **Fig. 58.20** — Forms of vaginal obstruction: imperforate hymen, transverse septum, vaginal agenesis (Yerkes and Rink, 2002).
- **Fig. 58.26** — MRI, 12-year-old with **Klippel-Feil and proximal vaginal + cervical atresia: no cervical impression** at the lowest point of the uterus.
- **Fig. 58.27** — Müllerian fusion anomalies: partial bicornuate, bicornuate, uterine duplication, complete uterine and vaginal duplication, duplication with unilateral imperforate vagina (Yerkes and Rink, 2002).
- **Fig. 58.28** — Genitogram: complete müllerian duplication (septate vagina, separate uteri).

- **Fig. 58.29** — HWWS: angiocatheter in the obstructed side.

Concept — Vaginal anomalies are sorted by **which primordium failed and whether a functioning uterus sits above the block**: a uterus above an obstruction is an emergency of drainage, an absent uterus is an elective problem of creating a vagina, and an absent cervix is the one situation where joining the two is dangerous.

ACQUIRED DISORDERS OF THE FEMALE EXTERNAL GENITALIA

Rhabdomyosarcoma

- Vaginal rhabdomyosarcoma: **grapelike cluster from the posterior vestibule**; mean age <2 years.
- **Best prognosis of all female genital tract primaries** — embryonal predominance + early detection from bleeding.
- **Biopsy → staging**: computed tomography (CT) of abdomen and chest + bone marrow biopsy → protocol stratification.
- Chemotherapy advances have steadily **reduced surgery** across the **Intergroup Rhabdomyosarcoma Study (IRS)** trials.
- Local resection may follow chemotherapy; **wide excision of the organ only for persistent or recurrent disease**.

PROPORTION UNDERGOING SURGERY ACROSS IRS TRIALS

IRS-I	IRS-II	IRS-III	IRS-IV
100%	70%	30%	13%

- The trend is the point: **chemotherapy has displaced extirpative surgery**.

Labial Adhesions

- Also **labial agglutination / synechia vulvae**; **commonest interlabial abnormality** (0.6–1.8%); mainly **first 2 years**, peak **13–23 months**.
- **Labia minora fuse from the posterior fourchette towards the clitoris**; distinguish from **labia majora fusion (DSD)**.
- **Not seen in newborns** (maternal oestrogen) → hypo-oestrogenism hypothesised, but **circulating oestradiol is not different** from controls → questioned. Oestradiol may aid **re-epithelialisation** (heparin-binding epidermal growth factor) — i.e., helps healing after separation.
- **Local irritation and trauma** are important triggers; rarely **sexual abuse** (haematoma, lacerations).
- Usually asymptomatic; may cause **post-void dribbling** (vaginal pooling), perineal irritation, contaminated urinalysis, difficult imaging.
- **Spontaneous resolution up to 80% within 1 year** → **observe if asymptomatic**.

LABIAL ADHESIONS — TREATMENT LADDER

Step	Detail
Observation	Asymptomatic
Topical oestrogen	Randomised trial: oestradiol superior to emollient ; conjugated oestrogen often effective; side effects breast budding, hyperpigmentation
Topical betamethasone 0.05%	Twice daily 4–6 weeks, 1–3 courses; alternative avoiding oestrogen effects
Technique	Cream directly on labia minora ; family performs pull-down several times daily

Step	Detail
Manual separation (office)	Eutectic mixture of local anaesthetics (EMLA) cream under Tegaderm occlusion → gentle pressure with lubricated probe
Surgical lysis (anaesthesia)	Dense adhesions ; oversewing with 7-0 chromic optional

- Recurrence is **similar after manual and surgical separation**; what determines it is **family compliance with daily separation**. Oxidised cellulose barrier: adults only.

Female Circumcision (Infibulation)

- Ritual **female genital mutilation (FGM)**: Africa (Horn of Africa through central Africa to Nigeria), Middle East, Muslim populations of Indonesia and Malaysia. Practised in 19th-century Britain for “epilepsy to lesbianism”; **now illegal in Western countries**.
- **~200 million females affected; ~3 million African girls annually**.
- Ethiopian data: associated with **ethnicity/religion and higher socioeconomic status**; **inversely related to maternal education**.
- Age: birth to pre-marriage; typically **4–10 years, commonest age 7**; usually **without anaesthesia** as a rite of passage. **Predates Islam — not religious**. **Deinfibulation** often before consummation.
- Range: **sunna** (clitoral prepuce excision) → complete vulval excision. **Pharaonic circumcision** = infibulation by sutures, thorns or twigs; **legs bound up to 40 days**.

FGM CLASSIFICATION — TOUBIA (1994), AS IN SOURCE, AND CURRENT WORLD HEALTH ORGANIZATION (WHO) TYPOLOGY

Toubia	Extent	WHO equivalent
Type I	Clitoris complete/partial removal	Type I (clitoridectomy)
Type II	Clitoris + part of labia minora	Type II (excision)
Type III	Clitoris + minora excised; medial majora raw, anterior two-thirds approximated	Type III (infibulation)
Type IV	As III with near-complete closure, pinhole at fourchette	Type III (infibulation)
—	—	Type IV: all other harmful non-medical procedures

- Trap: the **numbering differs between schemes** — in the WHO typology infibulation is **Type III**.
- Sequelae: immediate **tissue destruction and infection** (rectum, urethra); long-term **chronic pain, recurrent urinary tract and vaginal infection, dysmenorrhoea, dyspareunia, apareunia**; revision may be needed for intercourse and delivery.
- Complications scale with **degree of destruction**: urinary tract infection, **retained menses with endometriosis, vaginal calculus** from urinary stasis.
- Care is **individualised, respecting culture**; **educate on normal anatomy**; photographs or drawings in consent.
- **Simple midline scar division with subcuticular closure of each edge** → **high success** and satisfaction.
- **Clitoral reconstruction**: less predictable (glans/corpora injury); reduced pain and restored pleasure reported, but **limited by poor follow-up**.

Condyloma Acuminata

- **Rising incidence** in children; mostly **perianal**, occasionally vestibular.
- **High suspicion for sexual abuse**, but **perinatal transmission** is a real mechanism.
- Treatment choice by **age, site, severity**; variable success. **Spontaneous resolution in over half**.
- **Imiquimod** — immune response modifier via **Toll-like receptor 7**; effective in rapidly progressive unresectable disease. **Carbon dioxide laser**: straightforward for anogenital lesions, harder at the vestibule.
- Cause = **human papillomavirus (HPV)** → **vaccination for primary prevention**.

Inguinal Hernias

- **Female : male ≈ 1:5 to 1:10**. Usually isolated, but may be the **presenting sign of CAIS (~1%)** → **every phenotypic female with a hernia is assessed for genetic male status**.
- **Karyotype definitive but costly** (not routine); **pelvic ultrasonography for vaginal stripe** — also costly.
- **Vaginoscopy at hernia repair** shows the cervix; CAIS also has a **shorter vagina**.
- **Inguinal laparoscopy**: contralateral gonad and uterus; **round ligament uniformly absent in genetic males**.

KEY POINTS: ACQUIRED DISORDERS OF THE FEMALE GENITALIA

- Labial adhesions: **no treatment unless symptomatic**.
- FGM: **~200 million females** worldwide.
- Condylomata: may be **perinatally acquired**.
- Inguinal hernia in a girl → **consider CAIS**.

Figure legends

- **Fig. 58.32** — Recurrent labial adhesions **after early silver nitrate to the labial edges**: incision marked with skin scribe, edges oversewn with absorbable suture.
- **Fig. 58.33** — Infibulation: scarred labia majora, **pinhole opening** only (Gonzales and Bauer, 1999).

Concept — Acquired lesions of the girl's vulva carry a **safeguarding shadow** (hymenal injury, adhesions with trauma, condylomata) and a **hidden-diagnosis shadow** (hernia → CAIS); the examination must actively exclude both before the lesion is treated on its own terms.